

BIOMECHANICS (EPHE 341)

Gait Analysis

Everything the course has built, pointed at
one ordinary thing: walking

By the end of this lecture you should be able to...

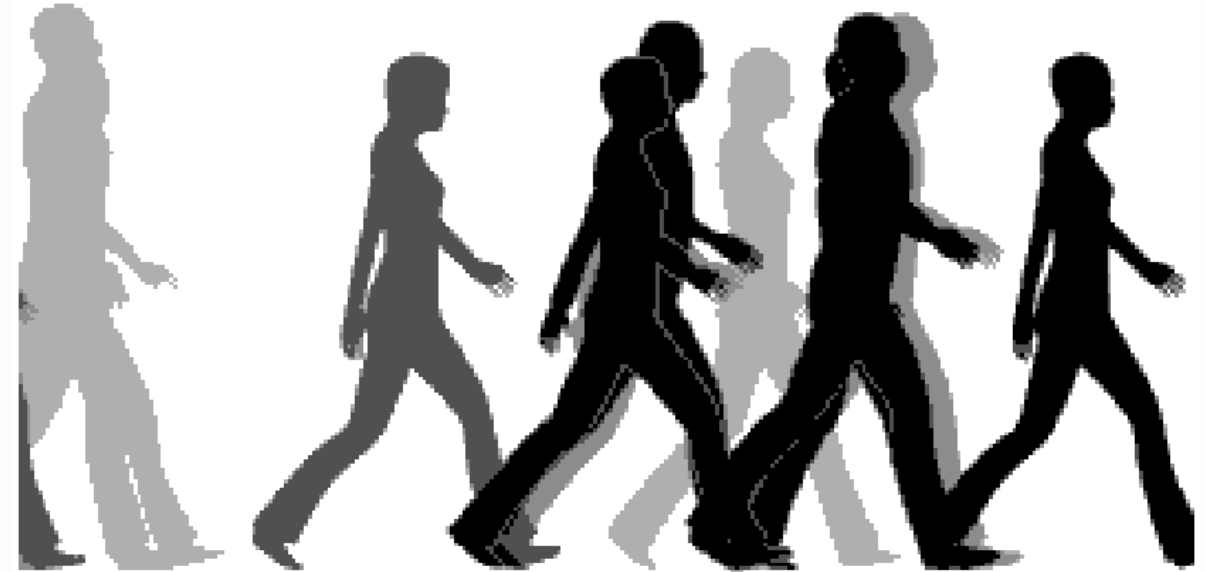
- **Divide** a stride into stance and swing and name the events that separate them
- **Describe** how a motion capture system turns reflective markers into joint angles
- **Link** a joint moment to the muscle group producing it and to the movement happening at the time
- **Account** for the inverted pendulum and for where the energy in walking actually goes
- **Define** the temporal and spatial variables, and say which ones change with speed, age and pathology
- **Read** the three components of the ground reaction force and the path of the centre of pressure
- **Explain** the support moment and why the body can trade one joint against another
- **Recognise** the common gait abnormalities and the problem each one is solving

What walking actually has to do

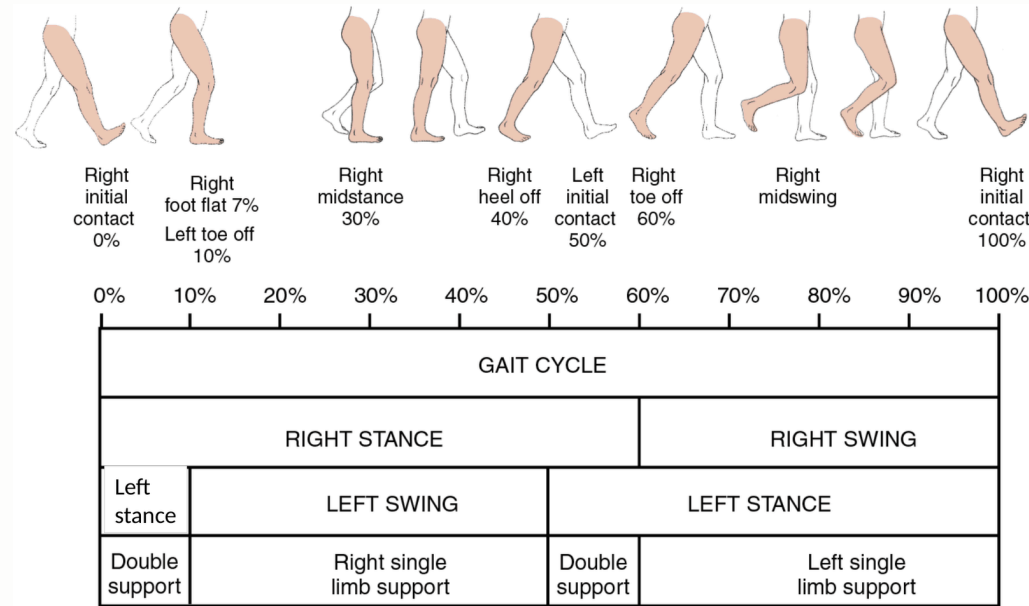
Gait repeats a basic sequence of limb motions that moves the body along a desired path while maintaining:

1. Support of the head, arms and trunk
2. Upright posture and balance of the body
3. Control of foot trajectory, for safe ground clearance and a controlled landing
4. Generation of mechanical energy to maintain or increase forward velocity
5. Absorption of mechanical energy, for shock absorption and stability, or to slow down

Five jobs at once, every step, with a body that is falling forwards most of the time. Understanding how it is normally done is what lets you say what has gone wrong when it is not.

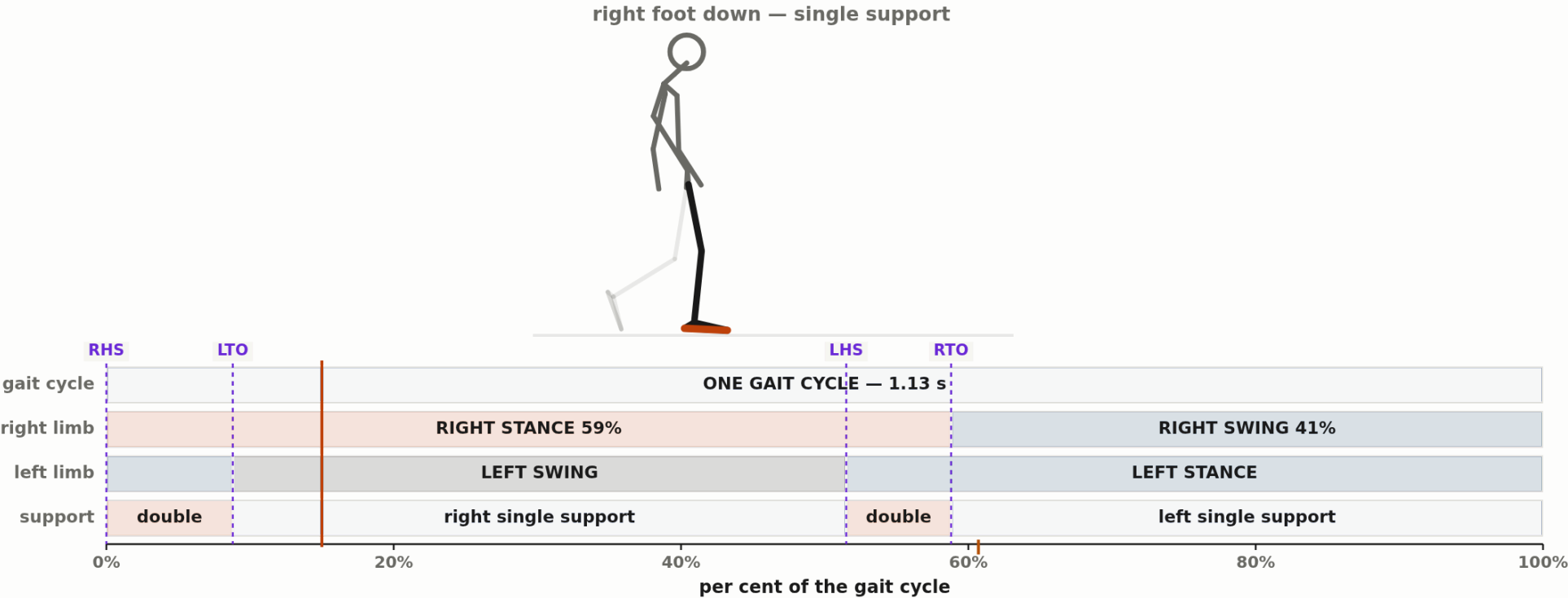


The gait cycle



- One **cycle**, or **stride**, runs from one event to the same event on the same foot — conventionally heel strike to heel strike.
- **Stance** is about 60% of it, **swing** the other 40%.
- The two limbs are half a cycle out of phase, so the left is in swing while the right is in stance.
- Both feet are down twice per cycle — **double support**, about 10% each time. It is the presence of double support that makes this walking rather than running.

The same diagram, from a recorded walk



Measured · Subject16 · 1.14 m/s · 106 steps/min · double support 16% of the cycle
events from the markers; the orange tick is right toe-off from the force plate, 1.9% later

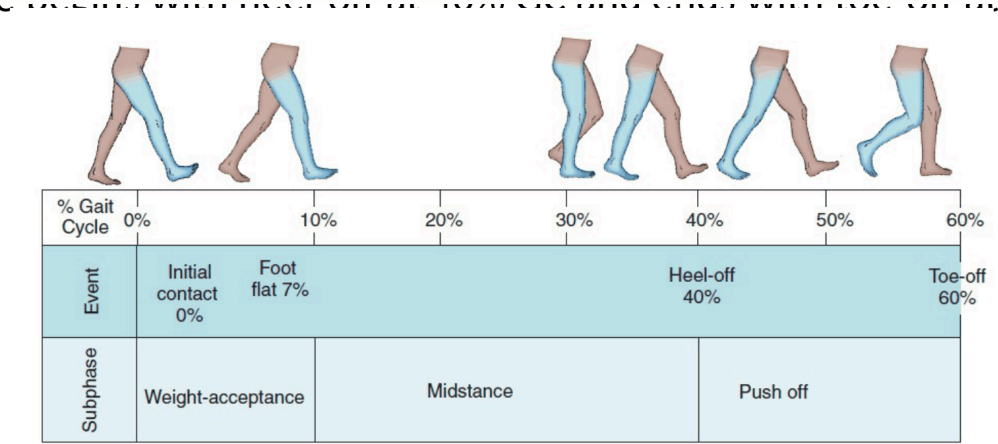
Gait cycle  15%

Subject16

Subject15

▶ Walk

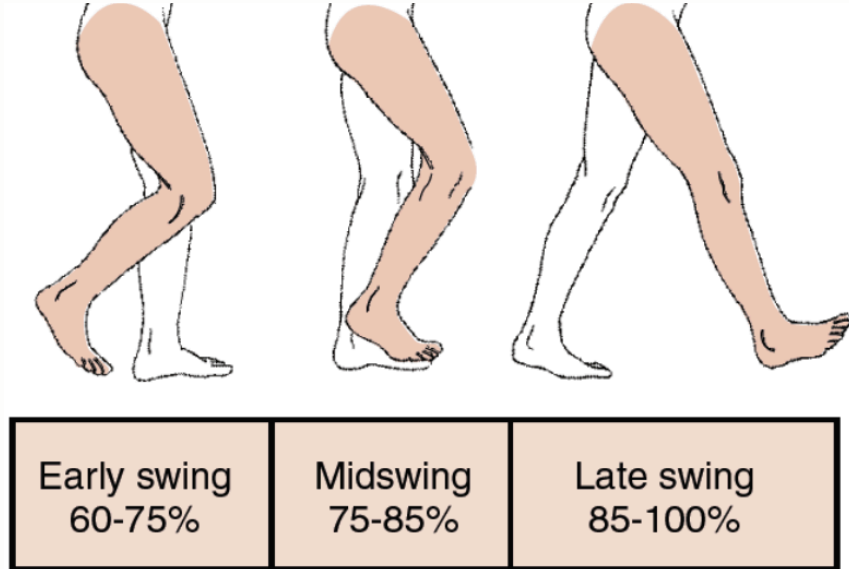
Stance, in four parts



1. **Weight acceptance** begins at initial contact and ends at foot flat, about 10% of the cycle. The limb takes the load and absorbs the impact.
2. **Midstance** begins at foot flat around 7–10% and ends at heel-off at about 40%. The body passes over the foot.
3. **Push-off** begins with heel-off at 40% and ends at toe-off at about 60%. This is where the leg does most of its positive work.

The names vary between textbooks — Perry's terminology splits stance into loading response, mid-stance, terminal stance and pre-swing. The events are the same; only the labels move. Read which scheme a paper is using before comparing numbers to it.

Swing, in three parts



1. **Early swing**, 60–75%, begins when the toe leaves the ground and continues until midswing. Also called initial swing, or the acceleration phase.
2. **Midswing**, 75–85%, is where the swinging leg passes directly beneath the body — the moment of least ground clearance.
3. **Late swing**, 85–100%, is the deceleration phase: the limb is being slowed in preparation for heel strike.

Midswing is the critical instant. The foot clears the floor by one or two centimetres and the knee is still partly flexed. Four of the five abnormalities at the end of this lecture exist to buy a little more clearance at exactly this moment.

The variables we actually measure

Temporal variables (time): stance time, single-limb and double-support time, swing time, stride and step time, cadence, and speed.

Spatial variables (distance): stride length, step length and width, and the degree of toe-out.

Each may be modified by age, sex, height, the distribution of mass in the body segments, joint mobility, muscle strength, clothing and footwear, habit, and psychological state.

That last list is why a single person's numbers mean little on their own, and why the useful comparisons are either **left against right** in the same person, or **before against after** an intervention.

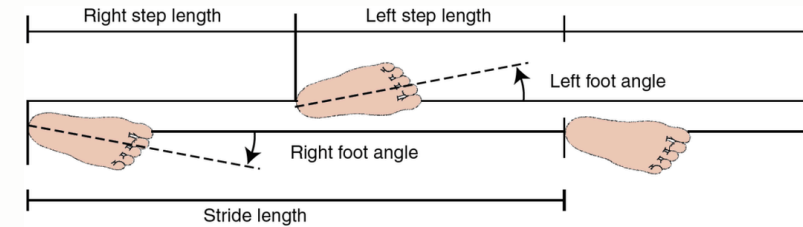


Figure 14-5 Stride length, step length, and width shown with foot angle placements. The midpoint of the heel is used as a point of reference for measuring step width.

Temporal variables — the times

Stance time — the time spent in stance by one limb during a gait cycle.

Single-support time — the time during which only that one limb is on the ground.

Double-support time — the time with both feet on the ground during one gait cycle, counting both occurrences.

- Double support is **increased** in elderly people and in those with balance disorders — it is the safest part of the cycle and a cautious walker spends longer in it.
- Double support **decreases as walking speed increases**, and reaches zero at the walk–run transition.

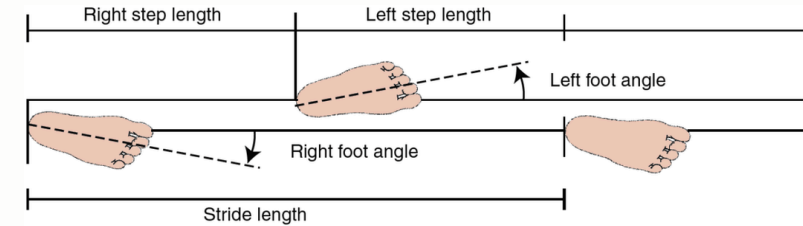


Figure 14-5 Stride length, step length, and width shown with foot angle placements. The midpoint of the heel is used as a point of reference for measuring step width.

Double support is the overlap between the two limbs' stance times — it does not appear on a footprint diagram at all, which is why both kinds of variable are needed.

Temporal variables — duration, cadence, velocity

Stride duration — the time to complete one stride. About one second in a normal adult.

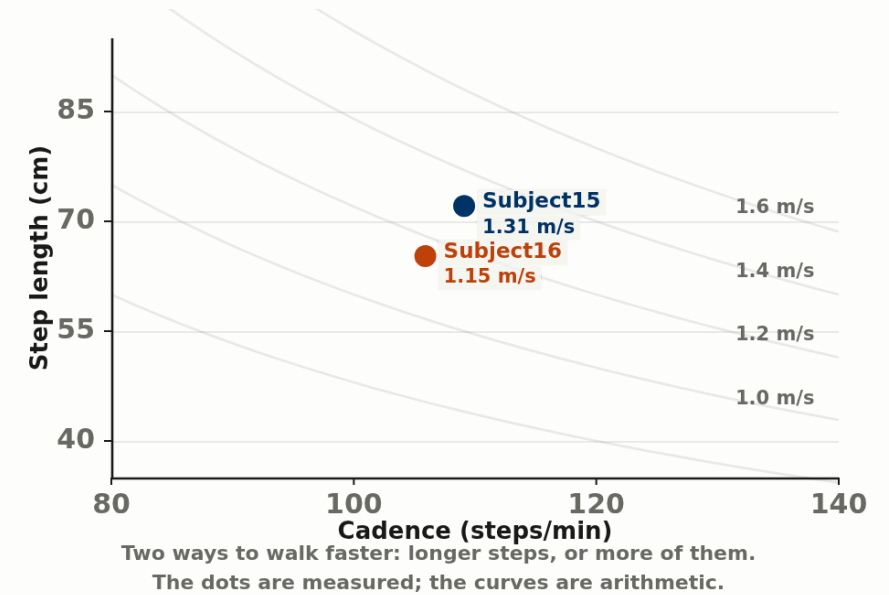
Step duration — the time spent on a single step. With weakness or pain in one limb, step duration decreases on the affected side and increases on the unaffected one.

Cadence — steps per unit time.

- A shorter step length gives a higher cadence at any given velocity.
- As cadence rises, double support shortens; approaching 180 steps per minute it disappears and running begins.
- Typical cadence is about **110 steps/min for men and 116 for women**.

Walking velocity — the rate of forward motion of the body.

$$\text{velocity} = \text{step length} \times \text{cadence}$$



Spatial variables

Stride length — the linear distance between two successive events accomplished by the *same* lower extremity.

- Affected by leg length, height, age, sex and more.
- Normalise by leg length or by height so that people of different sizes can be compared.
- Decreases in elderly people and increases with gait speed.

Step length — the linear distance between successive contacts of *opposite* extremities.

- Comparing right and left step lengths gives an indication of gait symmetry.
- Two step lengths make one stride, so they must sum to the stride length — a useful check on anyone's numbers, including your own.

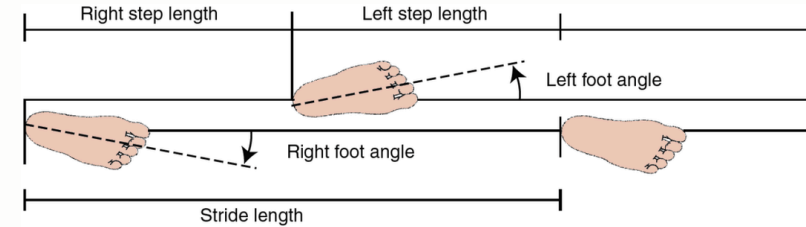


Figure 14-5 Stride length, step length, and width shown with foot angle placements. The midpoint of the heel is used as a point of reference for measuring step width.

Step width and toe-out

Step width — the width of the walking base, measured between the midpoint of one heel and the same point on the other.

- Increases when side-to-side stability is in demand: children, elderly people, anyone unsteady.
- In a normal adult population, mean step width is about **3.5 inches**, roughly 9 cm.

Degree of toe-out — the angle between the foot's line of progression and the line of travel.

- Normally about **7°** from the line of progression for each foot.
- Decreases as walking speed increases.

Note the care in the definition: the *midpoint of the heel*, not the back of it. A marker stuck on the back of a shoe is not that point, and the difference shows up in the numbers.

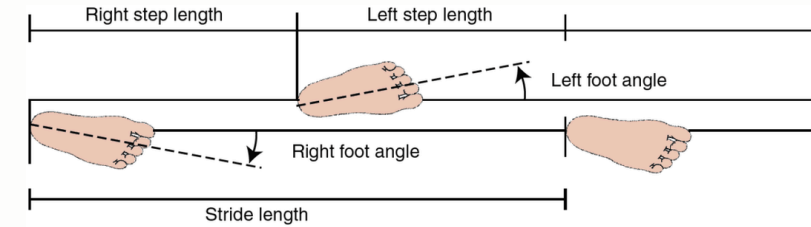


Figure 14-5 Stride length, step length, and width shown with foot angle placements. The midpoint of the heel is used as a point of reference for measuring step width.

The footprint diagram, measured



Subject16 · 161 cm · 1.14 m/s

stride length 130.5 cm

the authors' own value 130.3 cm — agrees to 0.2 cm

step length R 66.1 L 64.4 cm

they sum to 130.5 cm, which is the stride

step width 9.1 cm

the authors say 10.2 cm — a heel marker is not the midpoint of a heel, and the gap is the definition

toe-out R 4.5° L 6.0° (textbook ≈ 7°)

measured to the midpoint of the metatarsal heads

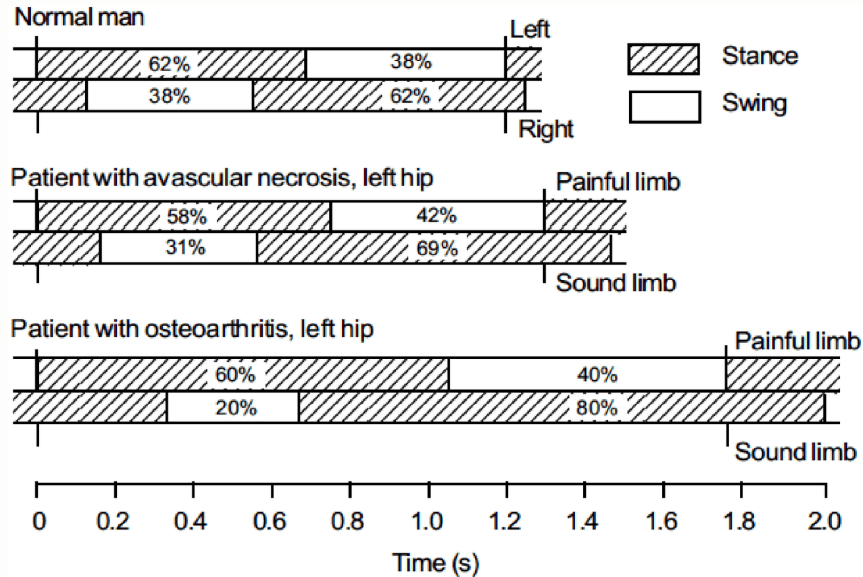
Subject16

Subject15

Why not the toe marker?

Stride length computed here from the heel markers lands within two millimetres of the authors' own published value, which is the check that the method works. Step width does **not** agree, by the same centimetre on both subjects — that is a difference of definition, not an error, and the figure prints both rather than hiding it.

Asymmetry

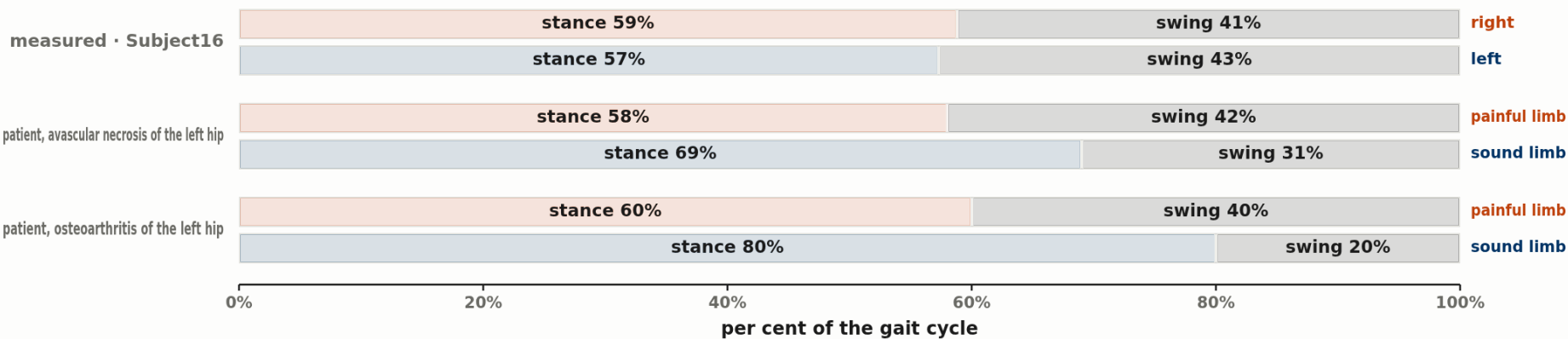


The clearest way to show a limp is to draw both limbs' stance and swing against the same clock.

- **Normal** — 62% stance and 38% swing, the same on both sides.
- **Avascular necrosis of the left hip** — the painful limb spends less time in stance (58%) and the sound limb more (69%).
- **Osteoarthritis of the left hip** — the same pattern, further exaggerated: 60% against 80%.

The pattern is general: a painful or weak limb is **unloaded sooner**, so its stance time falls and the other limb's rises. This is the antalgic gait, and the stance-time ratio is a usable measure of it.

How symmetric is a normal walk?



measured asymmetry 1.5 per cent — that is the noise floor a limp has to beat
the lower two rows are his figure, for comparison

Unload the right limb

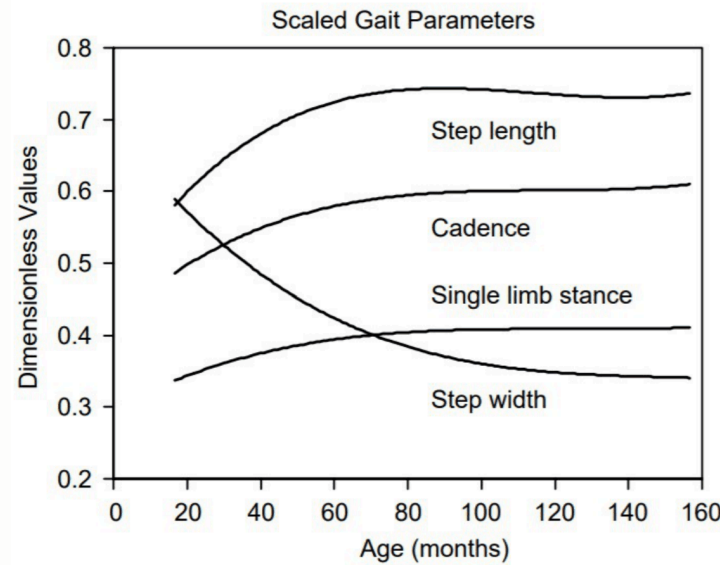
none

Subject16

Subject15

A healthy walk is close to symmetric and not quite — a per cent or two, repeatable, and nothing to do with pathology. Drag the slider and watch what **has** to move with it: the sound limb’s stance lengthens and both double-support periods shift, because the cycle still has to be a cycle. That is why gait variables are so correlated with one another.

Gait through childhood

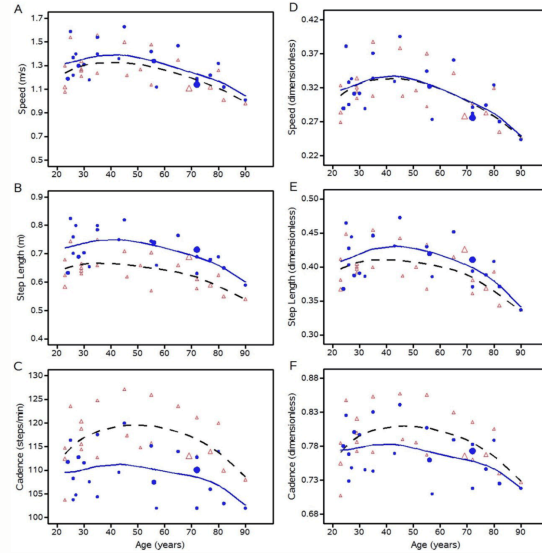


Scaled by leg length, the fundamental gait parameters change through the first six years of life and then stay very similar.

- **Step length** and **single-limb stance** rise — a child grows into longer, more confident steps.
- **Step width** falls steeply — the wide-based toddler gait narrows as balance improves.
- **Cadence** falls, because the legs are getting longer.

All four curves flatten by about 80 months — roughly seven years old. After that, a child's gait is an adult's gait at a smaller scale, which is why these plots are drawn **dimensionless**.

Age and sex in adults



Normalised to leg length, across adulthood:

- **Speed** declines at the same rate in both sexes.
- **Step length** decreases more in men.
- **Cadence** decreases more in women.

Two different routes to the same loss of speed. It is the kind of result that only shows up once the raw numbers have been normalised — the left-hand column of panels mostly shows how tall people are.

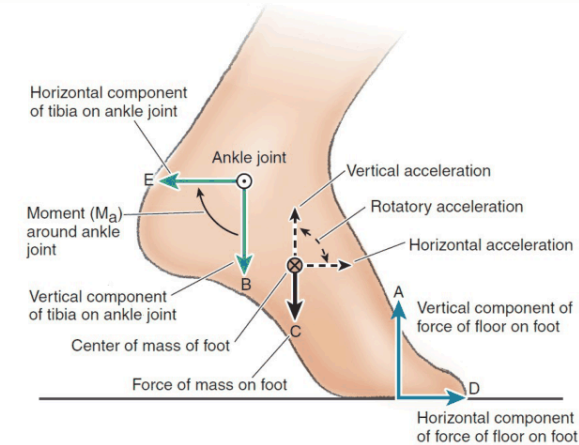
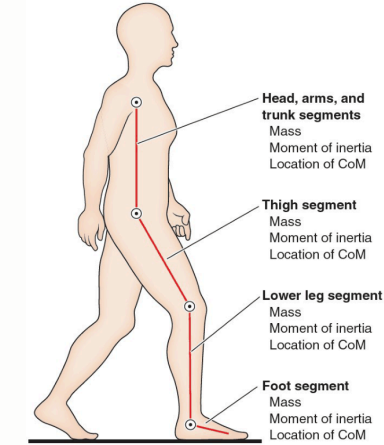
Beyond the basic parameters

The basic parameters are valuable, but they describe the *outcome* of walking. To ask what produced it we need the kinetics and kinematics of each joint.

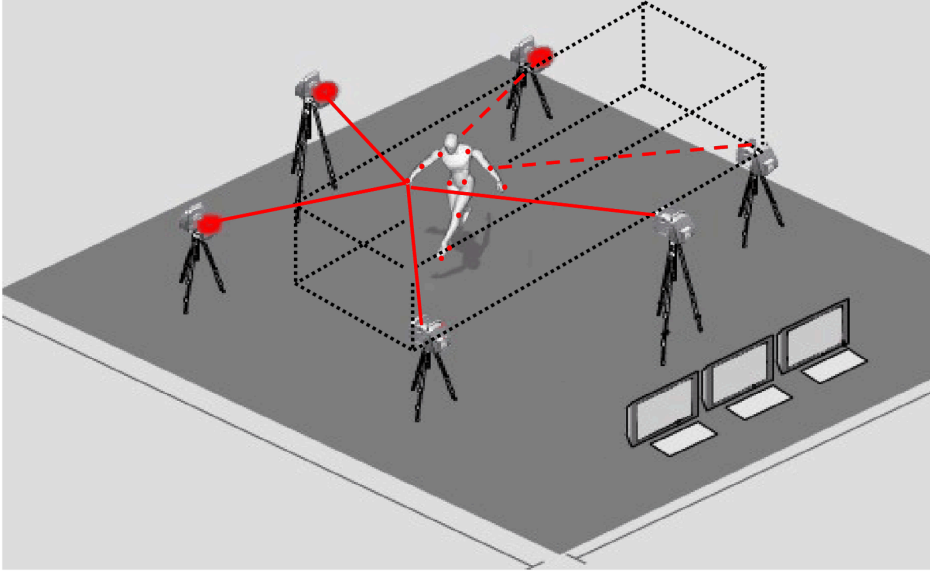
A **link-segment model** treats the body as a chain of rigid segments and uses inverse dynamics to find the forces and moments that must have acted at each joint. Start at the segment touching the ground and work upwards.

To do it, three things must be known for every instant:

- the **positions** of the segments through the cycle, from motion analysis;
- the **ground reaction force**, from a force plate;
- the **mass, centre of mass and moment of inertia** of each segment, from anthropometric tables.



Motion capture

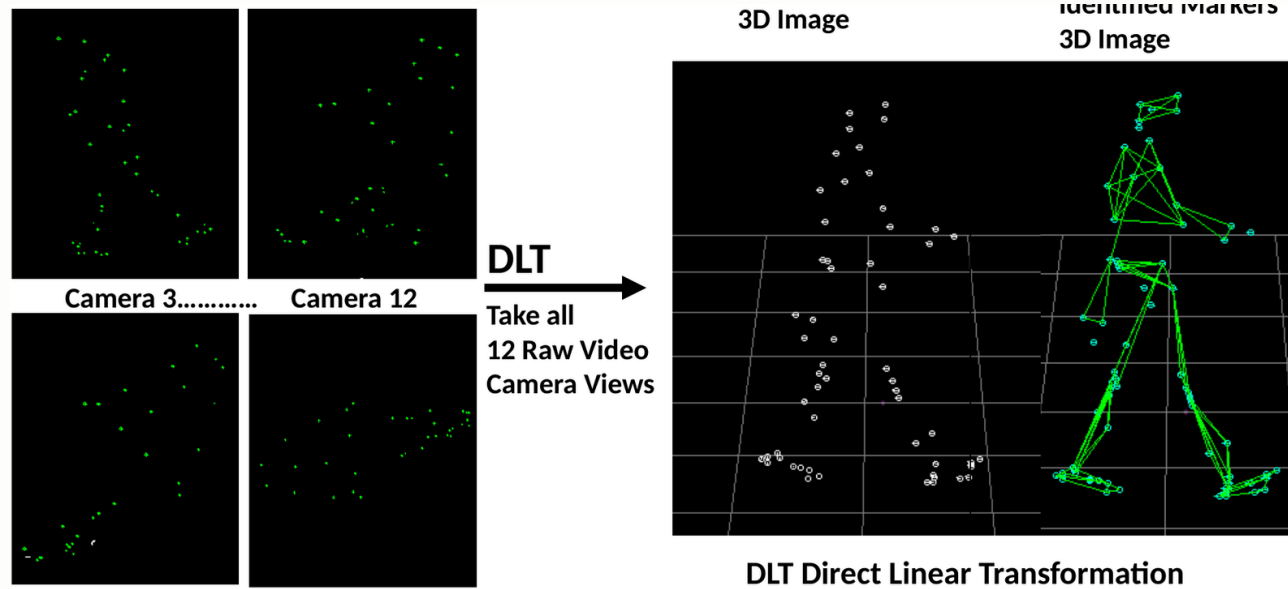


Reflective markers are placed on anatomical landmarks. Infrared cameras ring the volume, each flashing a ring of infrared light and recording what comes back.

Each camera sees a marker as a bright blob at a 2D image position. Given **at least two** cameras that can see the same marker, and a calibration that tells you where each camera is, the marker's position in space can be triangulated.

"At least two" is the theoretical minimum and never the practical one. Markers are hidden by the swinging arms, by the other leg, and by the body itself, so a system has eight to sixteen cameras to make sure two of them can always see every marker.

Twelve views, one point in space

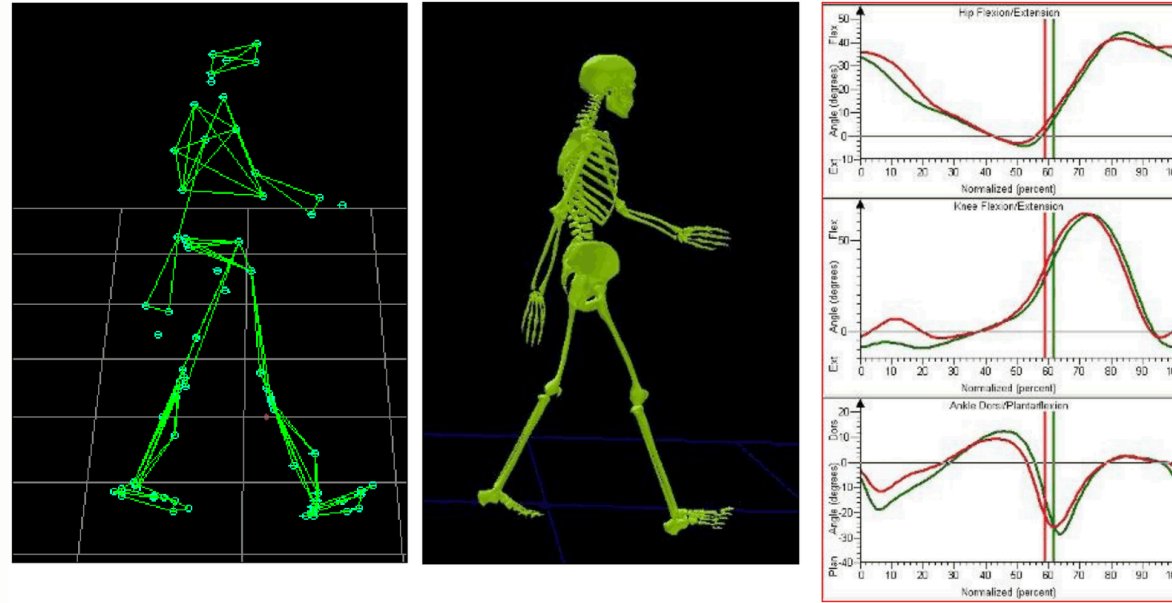


Each camera contributes a 2D view. The **direct linear transformation** (DLT) relates image coordinates to real-world coordinates through a set of calibration constants, found by waving a known object through the volume.

With every camera's view of a marker and every camera's DLT constants, the marker's 3D position follows by least squares — more cameras means a better-conditioned solution, not just redundancy.

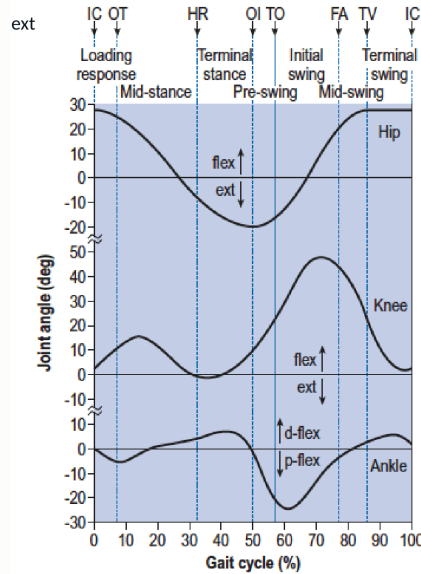
What comes out is an anonymous cloud of points. Turning that into "right lateral malleolus" is a separate step — **labelling** — and it is still the part of the process most likely to need a human.

From markers to joint angles



- Markers define each segment's **position and orientation**.
- Differentiating position gives velocity and acceleration, which inverse dynamics needs — and differentiation amplifies noise, so the data must be filtered first.
- A joint angle is the relative orientation of the two segments meeting at it — which means a joint angle is a *derived* quantity, two steps removed from what was measured.
- The output is normalised to 0–100% of the gait cycle so that strides of different durations can be averaged and compared.

The sagittal joint angles



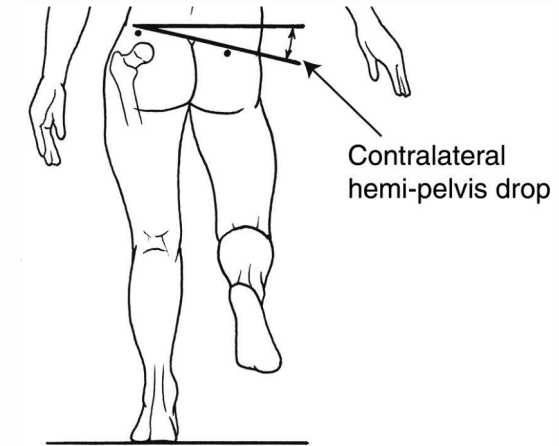
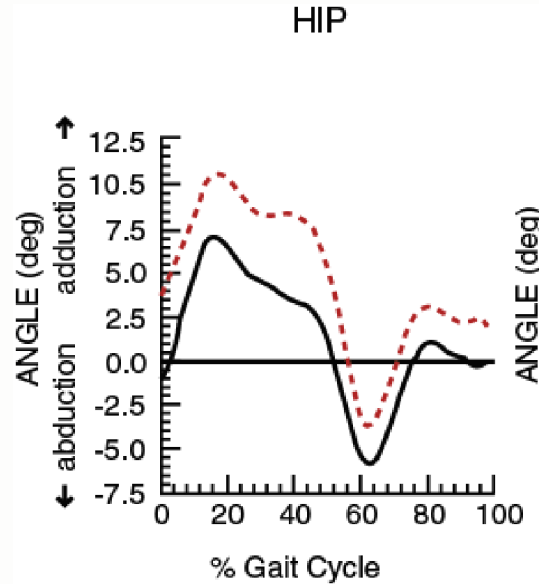
Hip — maximum flexion at initial contact; extends through stance to a maximum near heel-off at about 50%; flexes again through swing.

Knee — near 0° at initial contact and again just before heel-off, with a small flexion wave in loading response; maximum flexion of about **60°** at roughly 70% of the cycle, then extends ready for contact.

Ankle — maximum dorsiflexion at heel-off; maximum plantarflexion at toe-off, around 60%; then back to neutral for landing.

Each curve has **two** phases of motion per cycle, not one. The knee flexes twice — a small wave under load and a large one in swing — and mistaking one for the other is the commonest misreading of this graph.

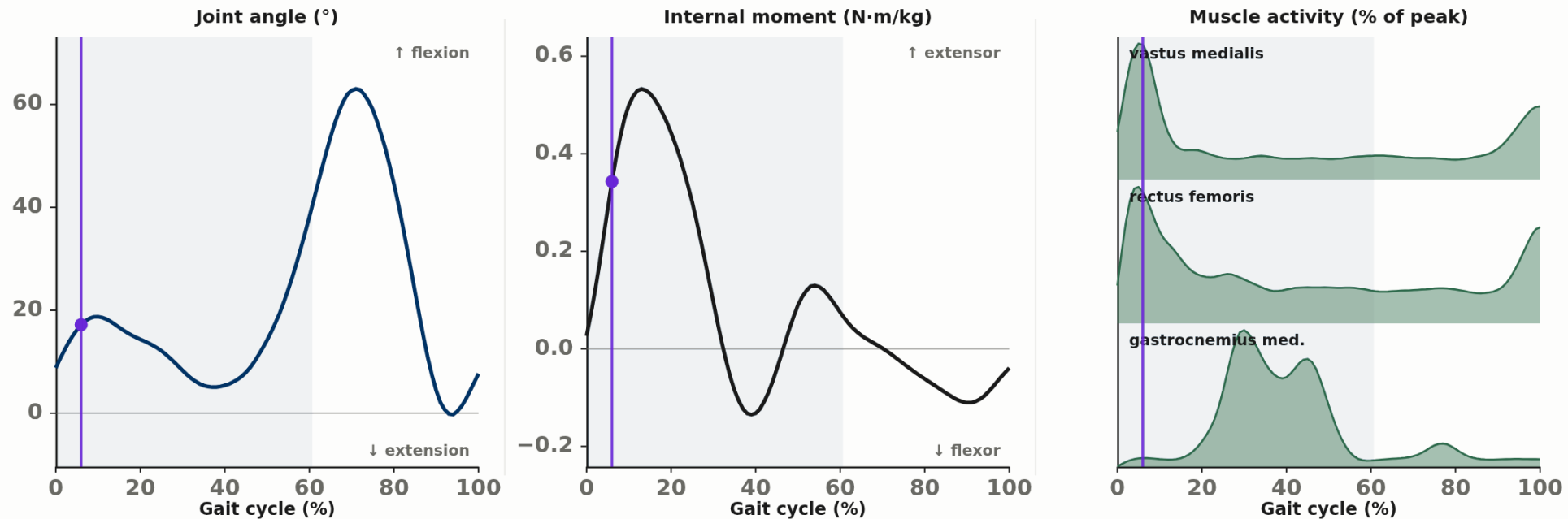
The frontal plane



During the first 20% of stance the pelvis on the **contralateral** side drops about 5°, which appears at the stance hip as **adduction**.

It is a controlled drop, not a collapse: the stance-side abductors are paying for it the whole time. Remove them and you have the Trendelenburg sign from the last lecture.

Angle, moment and muscle, one joint at a time



At 6% — knee moment 0.34 N·m/kg extensor, joint flexion
so that muscle group is working **ECCENTRICALLY** — lengthening under load, absorbing energy
Measured · Subject16 · shaded = stance

Gait cycle 6%

Hip

Knee

Ankle

Subject16

Subject15

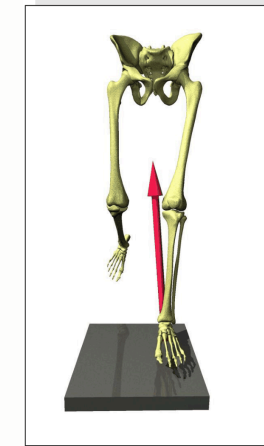
With **Knee** selected, put the playhead at **6%**: the knee is flexing, the moment is extensor and vastus medialis is on — the quadriceps are lengthening while they resist, lowering the body rather than lifting it. Move to **14%** and the same muscles are now shortening. Then try **Ankle** at **35%**, where the plantarflexors are lengthening at four times the moment — and at **50%**, where they have turned concentric.

What a force plate gives you

A force plate measures the resultant of everything the foot is doing to the ground, and by Newton's third law what the ground is doing to the foot. Three quantities come out of it:

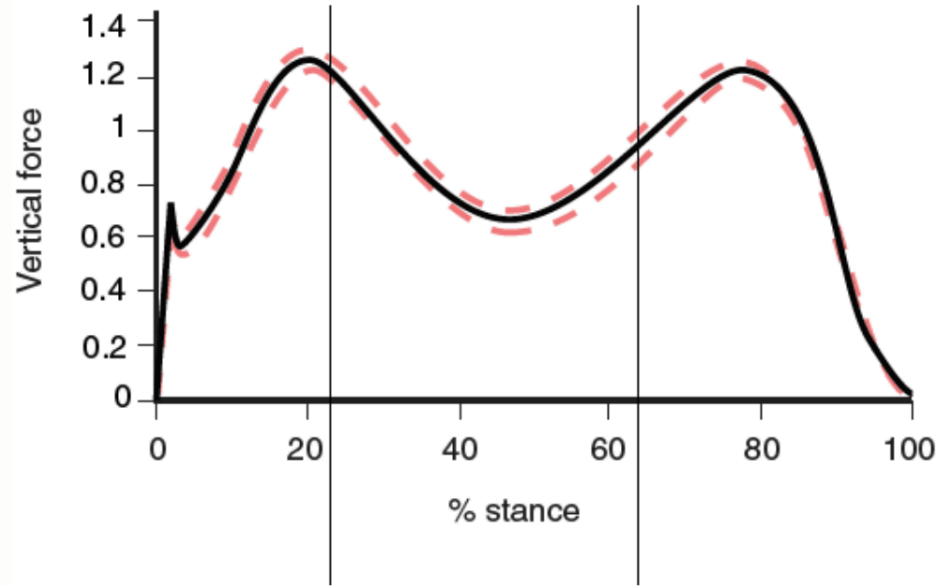
- the **ground reaction force**, as three orthogonal components — vertical, anteroposterior and mediolateral;
- the **centre of pressure**, the single point at which that resultant can be considered to act;
- the **free moment** about the vertical axis — the twist the foot applies to the floor.

It gives you **one resultant for the whole foot**. It cannot tell you that the heel is loaded and the forefoot is not; that is what a pressure mat is for, and the two are often confused.



Two plates set into the walkway, one per foot, sampled here at 960 Hz against the cameras' 60.

The vertical force

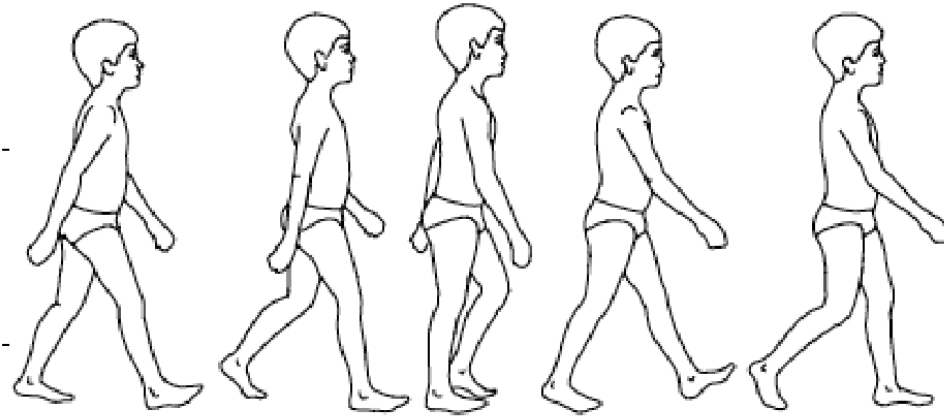


The vertical force has a characteristic shape: **two humps above body weight with a valley between them.**

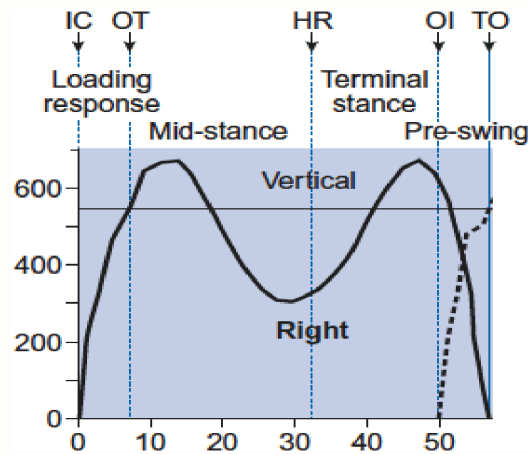
- The **first hump** is weight acceptance — the body's downward velocity is being arrested, which takes more than body weight.
- The **valley** at midstance is the body's centre of mass being accelerated *upward* less than gravity, as it passes over the foot. The knee unloads slightly.
- The **second hump** is push-off: the centre of mass is being accelerated upwards again, which once more takes more than body weight.

It is not a measure of effort. The humps are above body weight because the centre of mass is accelerating, and the valley is below it for the same reason with the sign reversed. $F = ma$, applied to the whole body.

Contact to midstance



Initial contact Loading response Mid stance Terminal stance Preswing

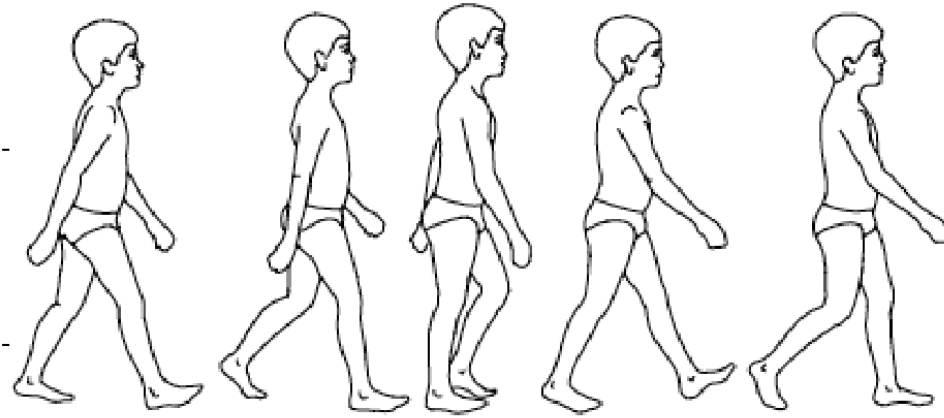


Initial contact (0%) — the instant the leading foot touches the ground. Preparation for loading response; heel strike or heel contact.

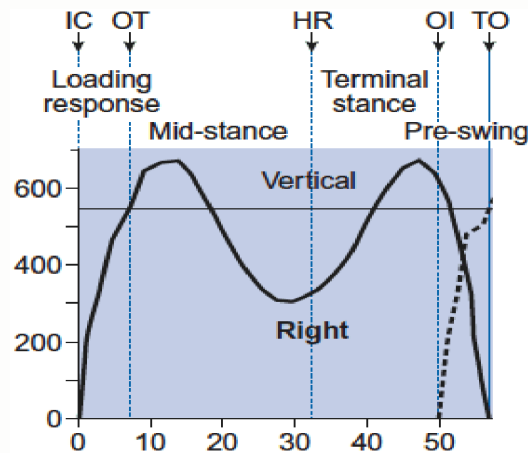
Loading response (0–10%) — body weight transferred onto the stance limb, foot comes into full contact, the vertical force rises to its first peak. This is the first period of double support.

Midstance (10–30%) — the opposite foot leaves the ground and body weight travels along the foot until it is aligned over the forefoot. The vertical force is on its descending limb. First half of single support.

Terminal stance to toe-off



Initial contact Loading response Mid-stance Terminal stance Preswing

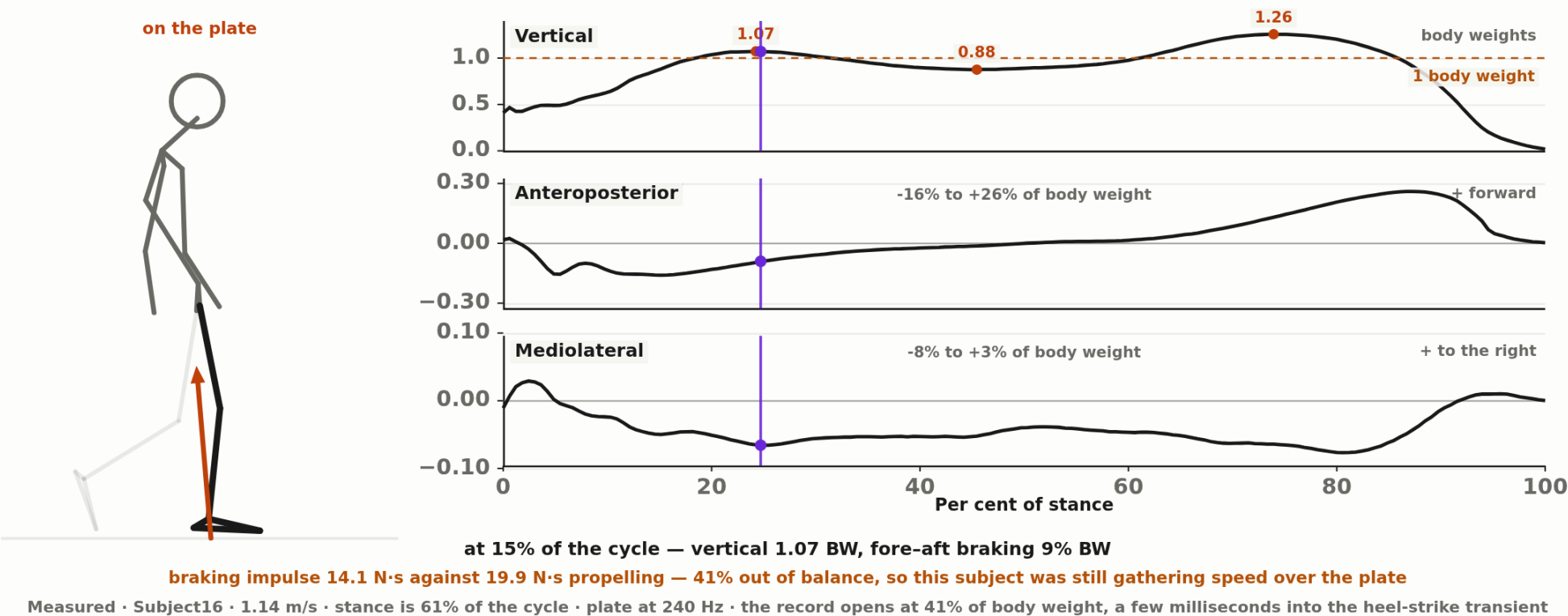


Terminal stance (30–50%) — begins with heel rise and ends when the opposite foot contacts the ground. Body weight moves ahead of the forefoot; the vertical force climbs to its second peak. Second half of single support.

Pre-swing (50–60%) — begins at opposite foot contact, ends at toe-off. The stance limb is unloaded and body weight transfers to the other limb. The descending limb of the second peak. Second period of double support.

Toe-off (about 60%) — the instant the foot leaves the ground. End of stance, beginning of swing.

All three components, measured



Gait cycle



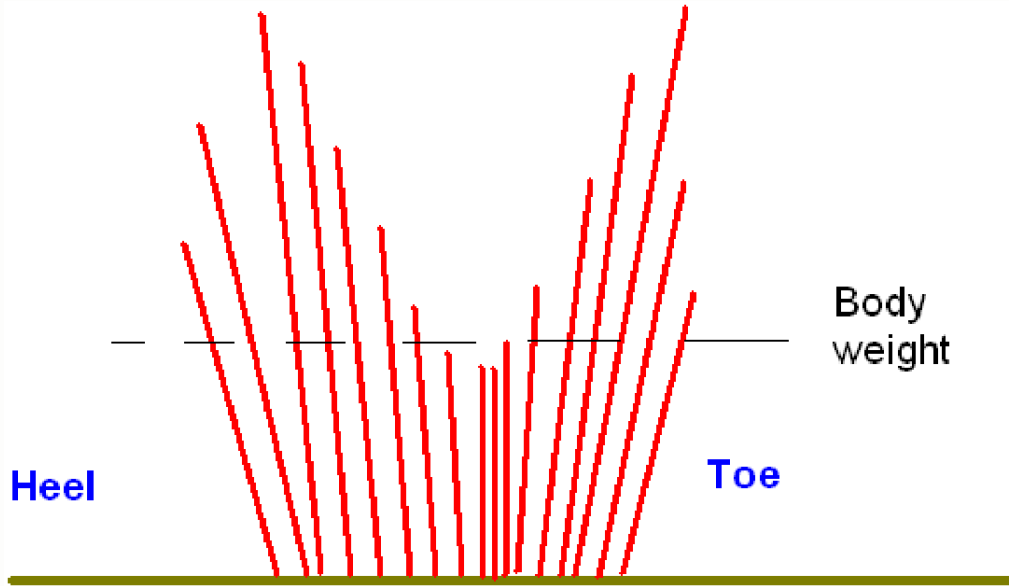
Subject16

Subject15

▶ Walk

Two humps and a valley, and the second hump is the larger. The fore-aft trace crosses zero at about **half of stance** — braking before, propelling after. Read the impulse line too: one of these strides was at a steady speed and one was not.

The force through the whole of stance

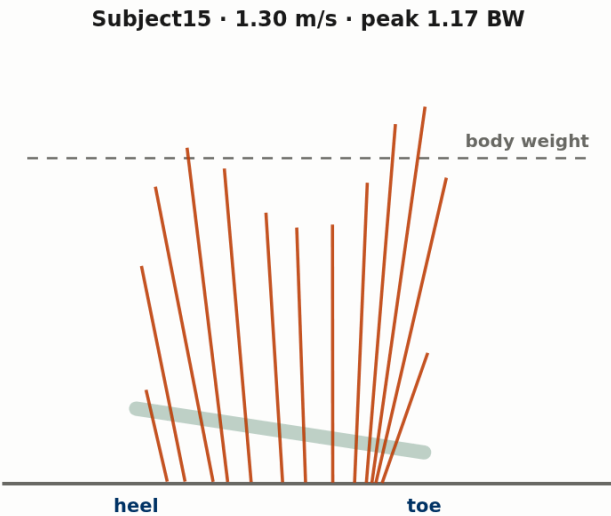
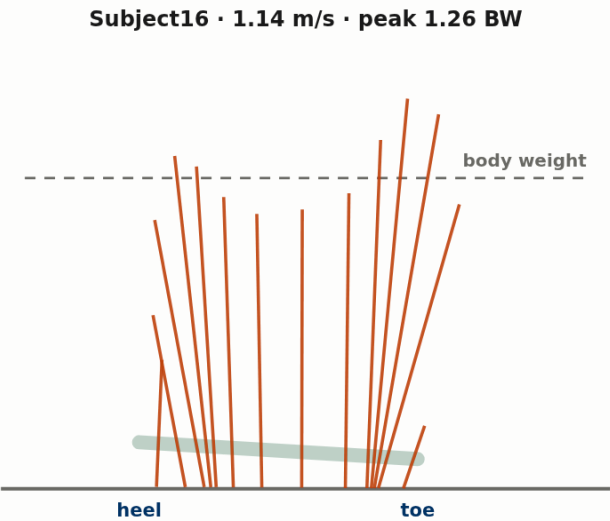


Draw the resultant force vector at intervals of about 50 ms through stance, each one from its own centre of pressure, and the whole of stance is on one picture.

- The leftmost line is the vector at the moment the foot touches the ground; each subsequent line is 50 ms later; the rightmost is at toe-off.
- The **length** of each line is the magnitude; the **lean** is the fore–aft component; the **foot** of each line is the centre of pressure.

Three variables on one figure, and the shape is so consistent between healthy people that it is sometimes called the **butterfly diagram** and read at a glance.

The butterfly, from measurement



13 force vectors each, one every 50 ms of stance
leaning back at the heel = braking · upright at midstance · leaning forward at the toe = propulsion
Measured · both subjects · force drawn with body weight at one foot length, which is how his figure is scaled

One vector every  50 ms

▶ One vector

At 50 ms these are his figure. Drop to 5 ms and the fans fill in — the shape is continuous and his picture is a sampling of it. The dashed line is body weight. Compare the two: same shape, different width, because the subject on the right walks faster and brakes harder.

Braking, then propulsion

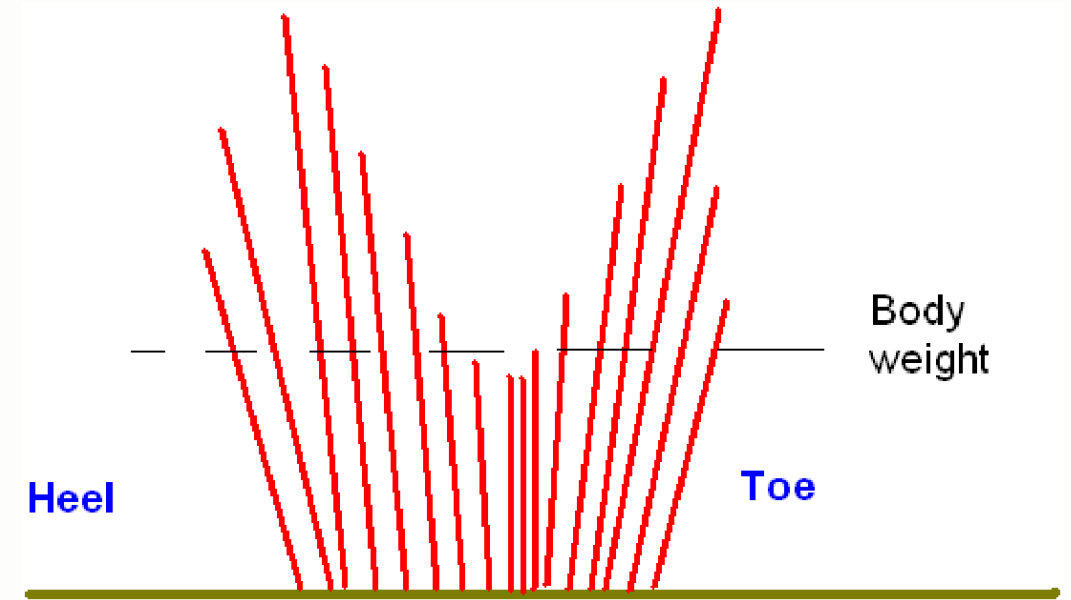
At initial contact the ground reaction force acts **diagonally backwards and upwards**, from the heel:

- the horizontal component acts *backwards*, braking the body;
- the vertical component is greater than body weight, arresting the controlled downward fall.

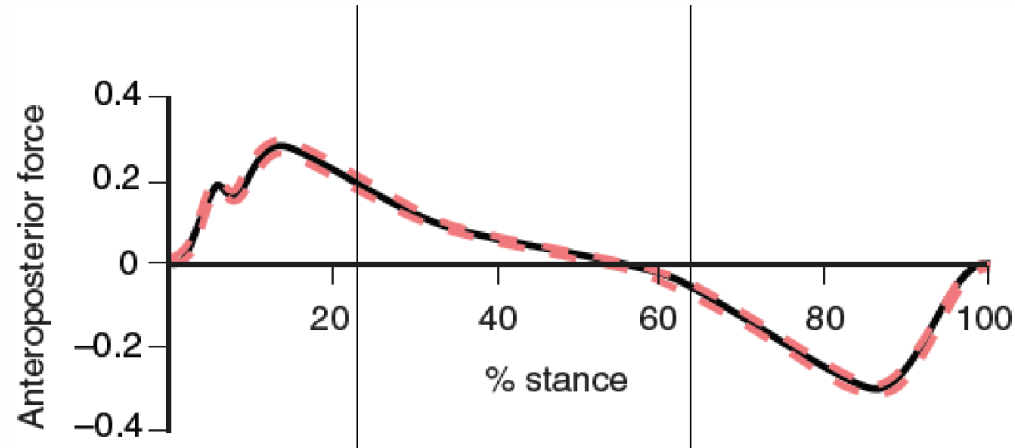
Through the middle of stance the point of action moves forward from the heel and the line of action becomes nearly vertical, so the braking effect disappears.

At the end of stance the force increases again, now acting **forwards and upwards**: the propulsion that keeps the body moving and stops it falling.

Over a stride at constant speed the braking and propulsive impulses must be **equal and opposite**. If they were not, you would be speeding up or slowing down — which, on a short overground walkway, is exactly what people are still doing.



The anteroposterior force

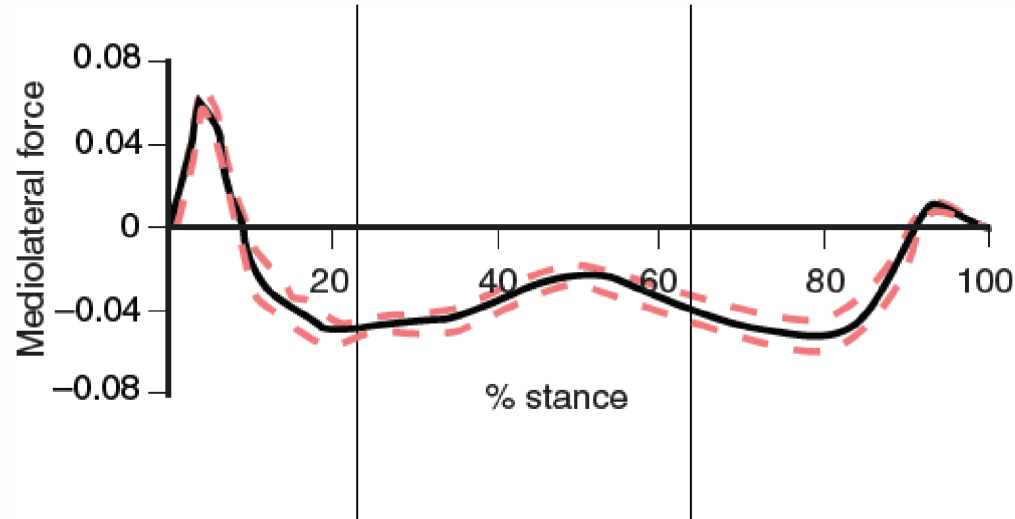


The anteroposterior force is directed **posteriorly** against the foot at initial contact, which is what stops the foot sliding forward. It reaches a maximum of about **20% of body weight**.

At midstance it passes through zero. Through the second half of stance it is directed **anteriorly**, enabling push-off.

Twenty per cent of body weight, horizontally, is also the friction the floor has to supply. A coefficient of friction below about 0.2 and the foot slips — which is why ice, polished floors and wet tiles are dangerous at exactly the two moments this trace is largest.

The mediolateral force



The mediolateral forces are **small in magnitude and variable across individuals**. Typical peaks are under 10% of body weight — an order of magnitude below the vertical force.

- A brief spike at contact as the foot lands and the body is caught laterally.
- A sustained, smaller component through single support, directed medially, keeping the body over the stance foot.

"Variable across individuals" is not a hedge. The two subjects in this lecture's data have mediolateral traces of *opposite* sign through much of stance, and both are perfectly normal walkers.

The centre of pressure

The single point on the foot at which the resultant surface pressure may be considered to act is the **centre of pressure**. It is the starting point of the ground reaction force vector, and therefore it is what sets every moment arm in the leg.

- It starts at the **heel** at initial contact.
- It moves **forward along the foot** through stance, quickly at first, then slowly through midstance.
- It ends under the **forefoot and toes** at toe-off.

It is a weighted average, so it can sit somewhere that is carrying no load at all — between two pressure peaks, or between the two feet during double support if the plate sees both.

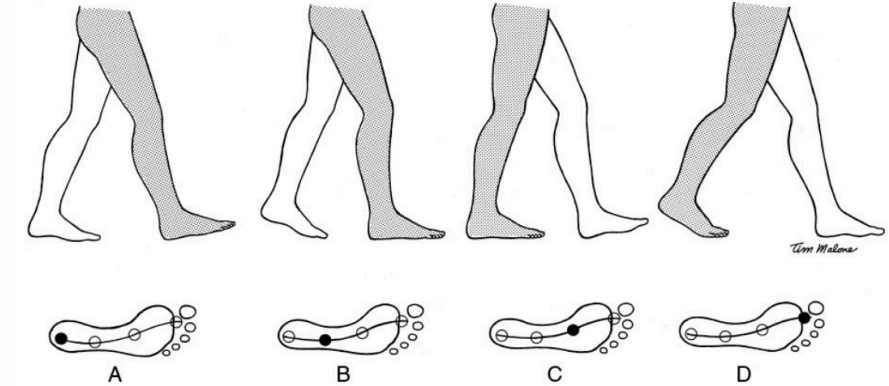
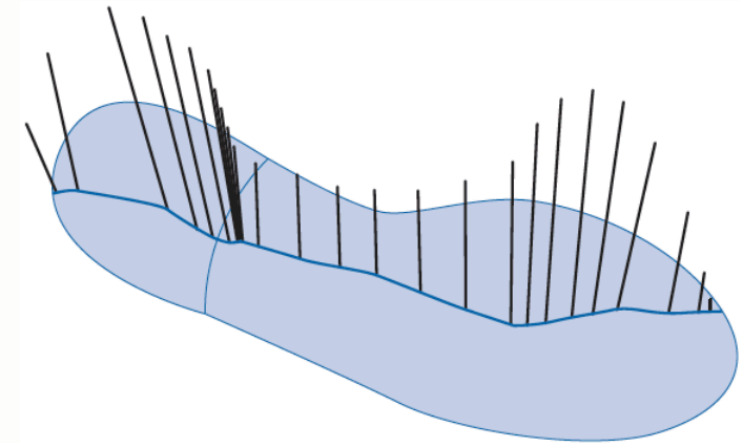
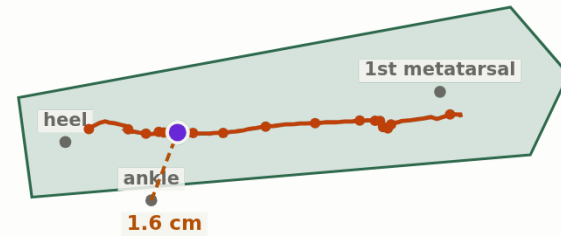


Figure 14-13 A center of pressure (CoP) pathway is shown by the position of the black dot at initial contact (A), at foot flat (B), just before heel-off (C), and just before toe-off (D).



The path under the foot

seen from above · travelling left to right · a dot every 40 ms



Subject16 · one stance phase

travels 22.9 cm along the foot

the heel-to-metatarsal distance is 23.3 cm,
so it ends ahead of that, under the toes

and 1.2 cm across

the lateral excursion is where most of the difference
between people lives

at 20% of the cycle

ankle moment arm 1.6 cm
× 530 N of vertical force

Gait cycle  20%

Subject16

Subject15

Ankle moment arm

The dots are evenly spaced in **time**, so where they bunch together the centre of pressure is moving slowly — that is midstance, the foot flat, the body rolling over. With the moment arm on, the distance from the centre of pressure to the ankle is the plantarflexors' lever, and you can watch it grow.

External and internal moments

External moments are produced about the joints by forces from outside the body — gravity and the ground reaction force.

Internal moments are produced by muscles, joint capsules and ligaments, and counteract them.

At initial contact, for instance, the ground reaction force passes behind the ankle, which tends to *plantarflex* it. The anterior tibial muscles answer with an internal *dorsiflexor* moment, and the foot is lowered under control instead of slapping down.

This is the angular kinetics material, and the convention is the one that catches people out: a gait graph labelled "dorsiflexor moment" means the **ground** was trying to plantarflex. The curve names the muscle group, not the direction of the external force.

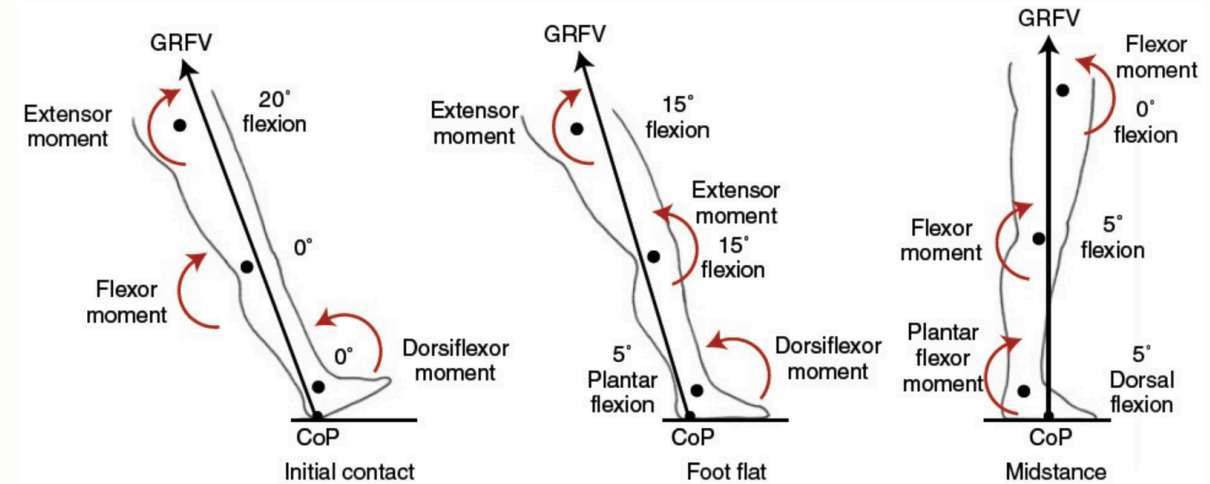


Figure 14-11 Diagram of joint positions with center of pressure (CoP), ground reaction force vector (GRFV), and internal net moments of force for gait cycle events of initial contact, foot flat, and midstance.

Which side of the joint, through stance

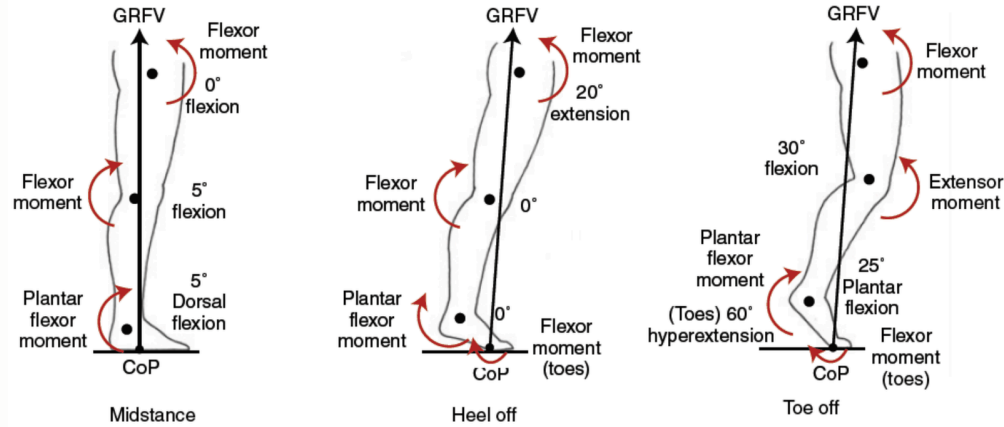
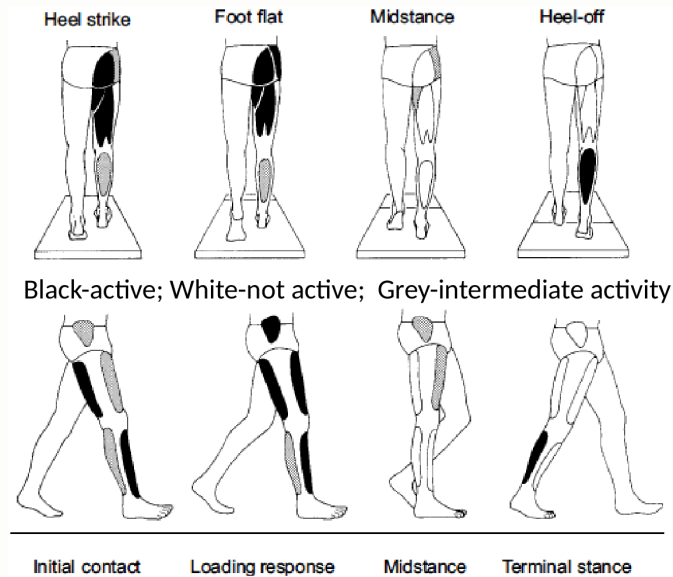


Figure 14-12 Diagram of joint positions with center of pressure (CoP), ground reaction force vector (GRFV), and internal net moments of force for gait cycle events of midstance, heel-off, and toe-off.

- At **initial contact** the vector passes behind the ankle — tending to plantarflex it — but in front of the knee and hip, tending to extend them.
- In **early stance** it swings behind the knee, tending to flex it, and the quadriceps answer. It moves ahead of the knee again by late stance.
- From about **midstance** it passes behind the hip and stays there until swing begins.
- At the ankle it moves steadily further in front of the joint as the centre of pressure runs forward, which is what builds the plantarflexor moment to its peak.

Muscle activity in stance

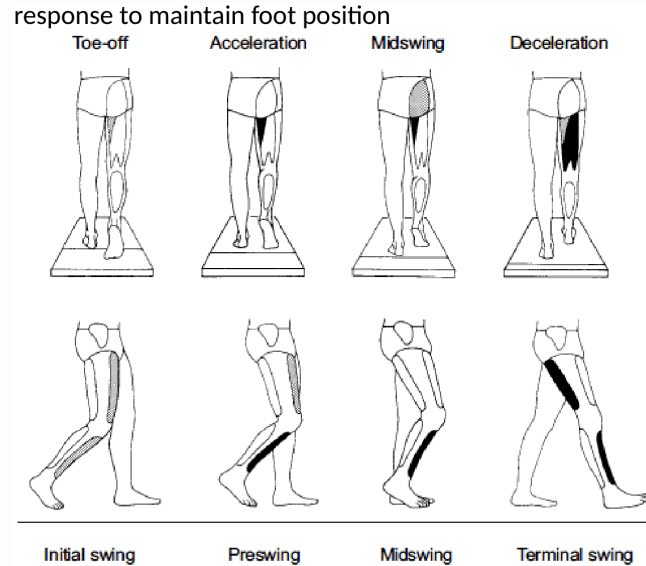


The timing of peak muscle activity tells you what each muscle is for.

- **Glutes** — hip extension during loading response.
- **Quadriceps** — knee extension during loading response.
- **Calves** — ankle plantarflexion during heel rise in terminal stance.

Read "extension" here as the *moment*, not the movement. The quadriceps are producing an extensor moment while the knee is **flexing** — they are lengthening under load, lowering the body rather than lifting it.

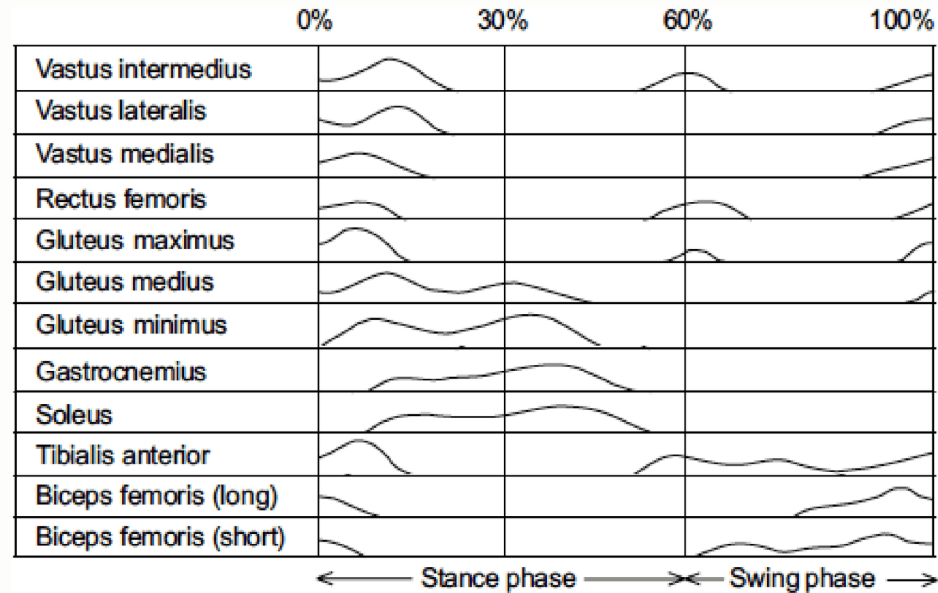
Muscle activity in swing



- **Iliopsoas** — hip flexion through pre-swing and initial swing, which is what starts the limb forward.
- **Hamstrings** — hip extension to *decelerate* the limb through terminal swing.
- **Tibialis anterior** — ankle dorsiflexion through swing to allow foot clearance, and again at loading response to lower the foot under control.

Tibialis anterior is the only one of the three that works twice per cycle, and it is doing a different job each time: **clearance** in swing, **controlled lowering** at contact.

The whole timing chart

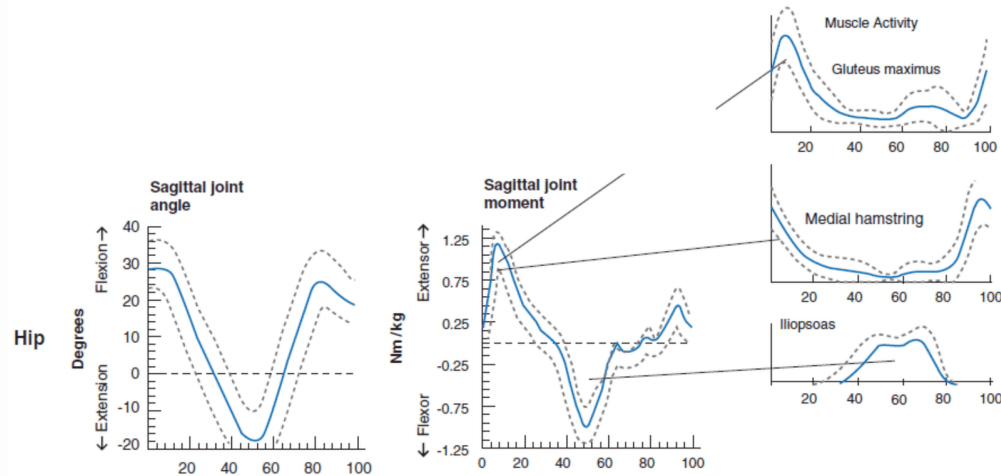


Twelve muscles, one cycle. Three things are worth reading off it:

- Most muscles are **off** most of the time. Walking is cheap because the muscles work in short bursts.
- Activity clusters at the **transitions** — loading response and the stance-to-swing handover — not in the middle of stance.
- Muscles crossing two joints (rectus femoris, gastrocnemius, the hamstrings) have more complicated patterns than one-joint muscles, because they are answering to two joints at once.

What a chart like this cannot show is the **deep** muscles. Iliopsoas appears on his figure and no surface electrode will ever record it.

Linking the moment to the muscle — the hip

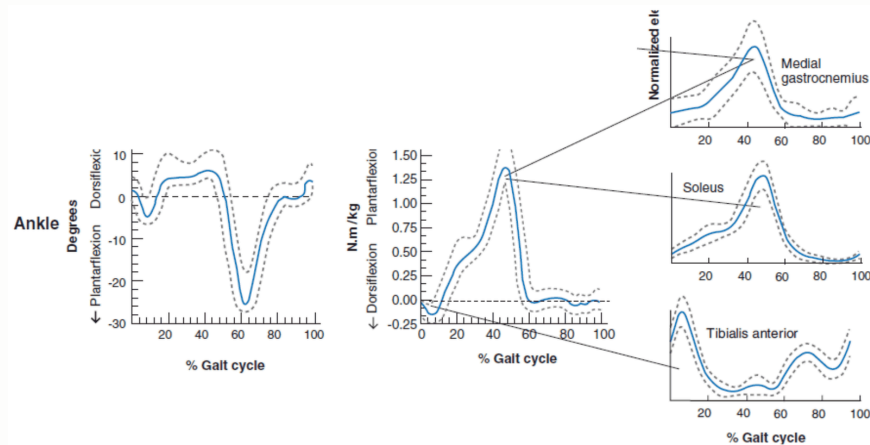
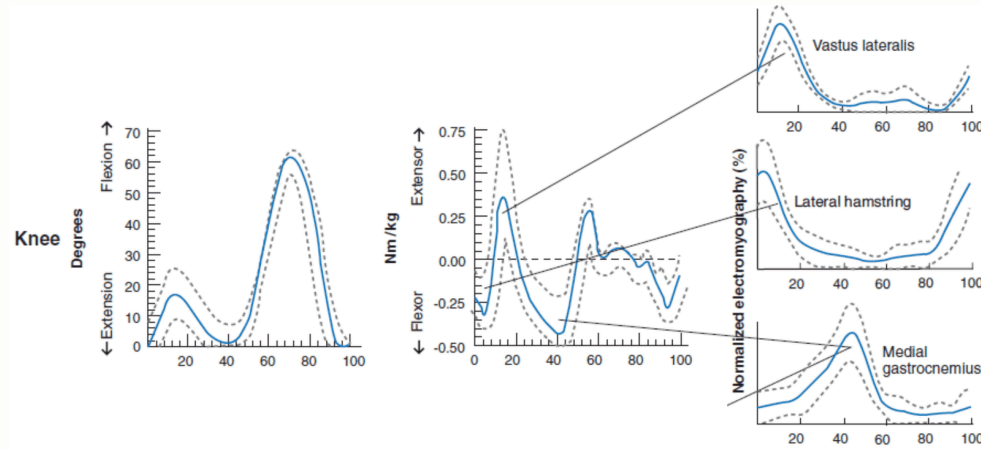


Internal joint moments can be linked to the muscles producing them.

- **Gluteus maximus and the hamstrings** support the extensor moment during weight acceptance.
- **Iliopsoas** supports the flexor moment through pre-swing and initial swing.

Three panels — angle, moment, muscle — are the minimum needed to say anything useful about a joint. The moment alone is ambiguous; with the angle it tells you concentric or eccentric; with the EMG it tells you which muscle.

...the knee, and the ankle

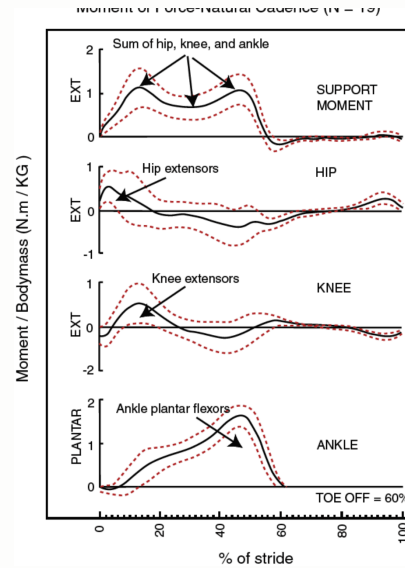


Knee. Vastus lateralis supports the extensor moment during mid stance; medial gastrocnemius supports the flexor moment during terminal stance — it crosses the knee as well as the ankle.

Ankle. Tibialis anterior supports the dorsiflexor moment during weight acceptance; soleus and medial gastrocnemius support the plantarflexor moment through terminal stance.

Gastrocnemius appears on both figures, producing a plantarflexor moment at the ankle and a flexor moment at the knee from the *same* contraction. A two-joint muscle cannot choose.

The support moment

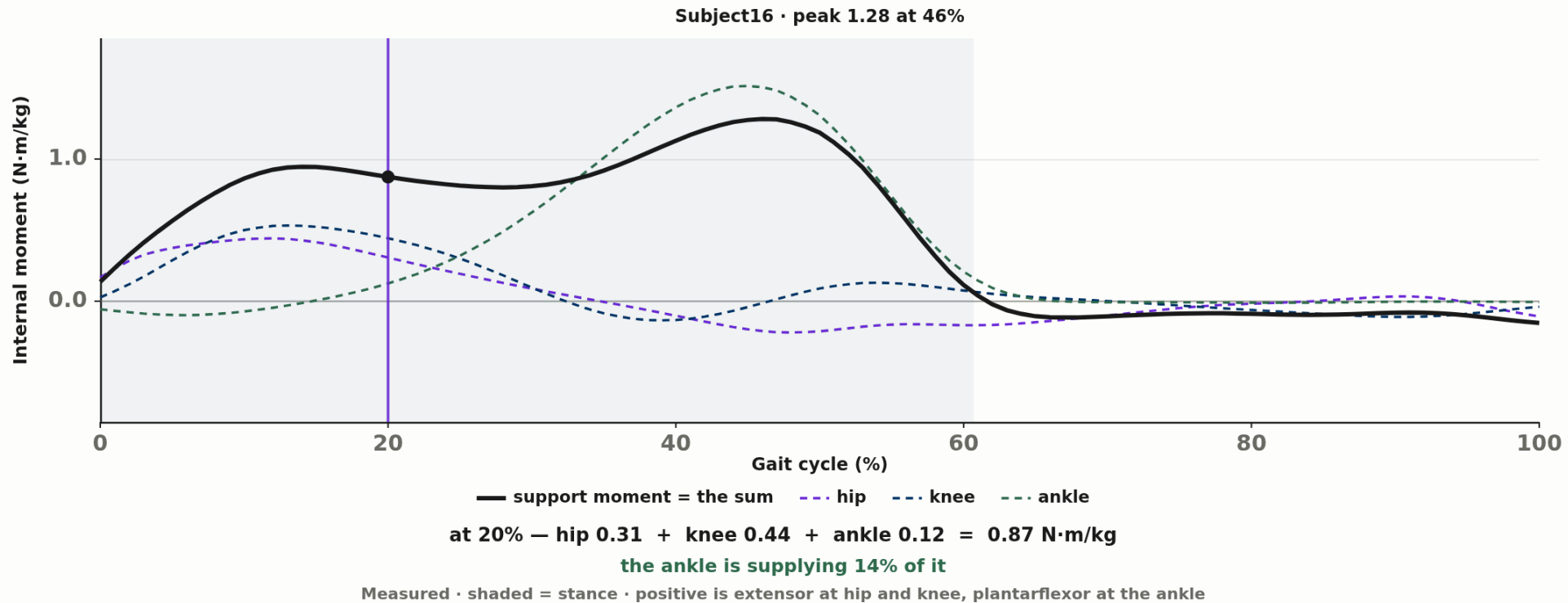


The **support moment** is the sum of the hip, knee and ankle moments, with extension positive at all three. It stays **extensor** through stance, and that is what keeps the leg from collapsing.

- Hip extension provides all of the early positive moment, joined by a knee extensor moment.
- As stance proceeds the ankle plantarflexors take over, until they are *the only support* through most of late stance.

$$M_s = M_{\text{hip}} + M_{\text{knee}} + M_{\text{ankle}}$$

Three joints, one total



Gait cycle 20%

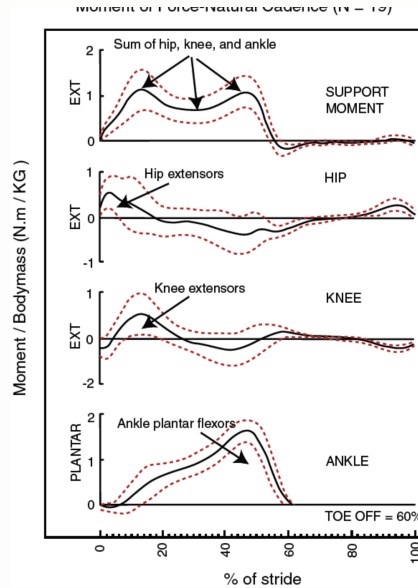
Subject16

Subject15

Both at once

The sum stays **extensor** through stance in both subjects — that is what keeps the leg from folding. Watch the handover: early on the hip and knee carry it, and by forty per cent the ankle is supplying more than the total, because the hip has gone flexor underneath it. Press **Both at once**: these two make the same support in visibly different ways, which is his point exactly.

Trading one joint against another



The support moment changes from a net extensor to a net flexor moment in late stance, which is what initiates swing. A small net extensor moment reappears in late swing, positioning the limb for contact.

As long as the moments add up to be extensors, the body can vary how it accomplishes its support. For example:

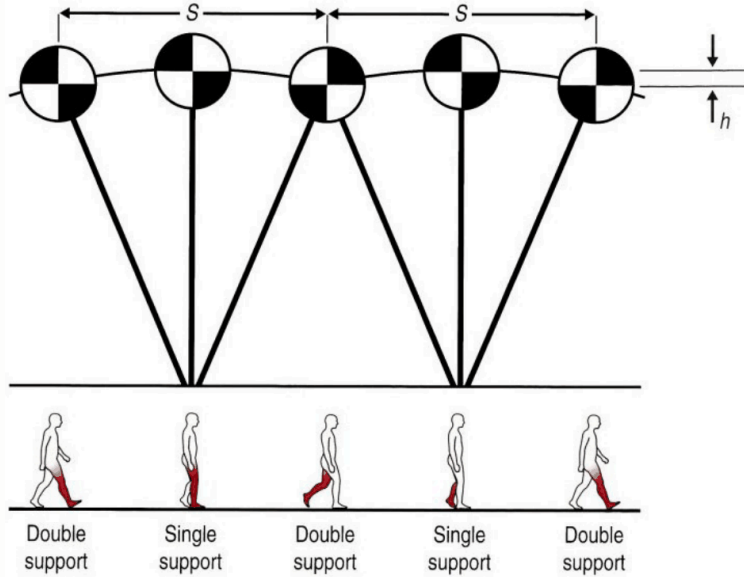
- if the ankle plantarflexors are weak, the hip extensors and the knee extensors can compensate;
- if contraction of the knee extensors causes pain, the hip extensors and the plantarflexors can contract harder.

This is why a single joint's moment is a poor clinical measure and the sum is a good one. The compensation is the finding.

Models, and what walking costs

Why a stiff leg and a falling body turn out to be a good idea

The inverted pendulum



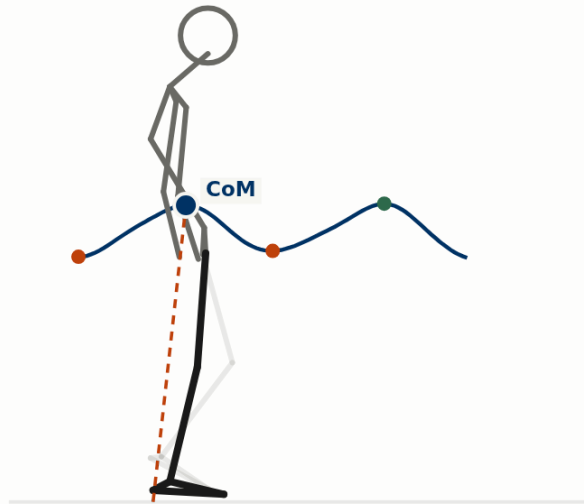
The mechanical analogy for walking is an **inverted pendulum**: during single-limb support the body vaults over a nearly straight limb.

- The centre of mass is **highest at midstance** and lowest at each double support — the opposite of what intuition suggests.
- Rising costs potential energy and slows the body; falling returns it and speeds the body back up.
- Potential and kinetic energy are therefore **out of phase**, and the exchange between them is nearly free.

A pendulum exchanges energy without any muscle doing work. That exchange is why walking at a comfortable speed is so cheap, and why it has an optimal speed at all.

Is a person actually a pendulum?

centre of mass path · vertical scale $\times 5$, or you could not see it

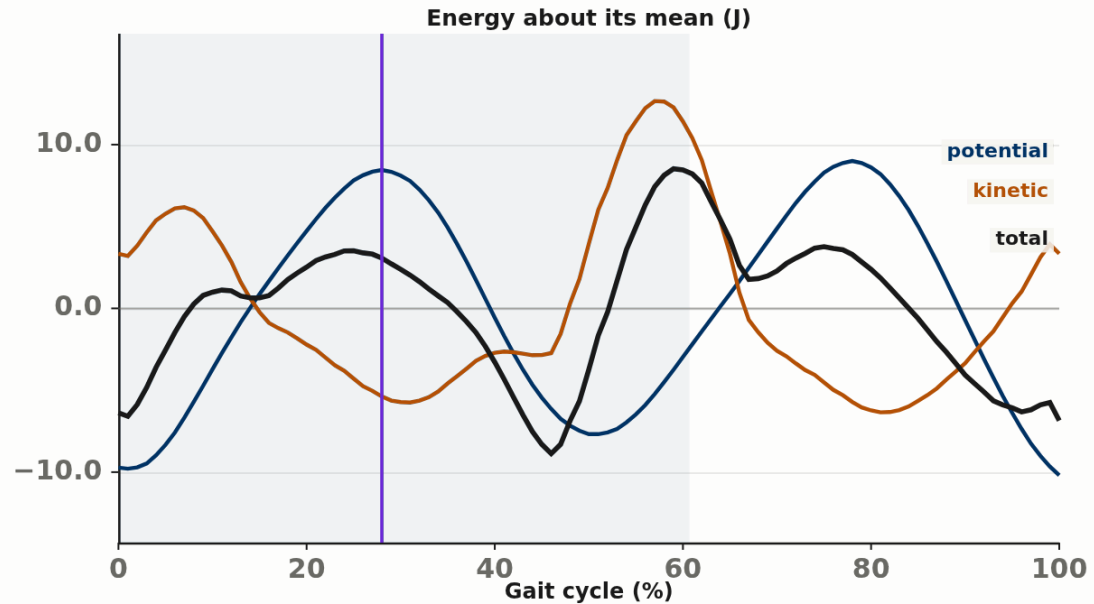


centre of mass rises 3.6 cm, twice per stride — highest at 28% and 78%, lowest at the two double supports

pendulum recovery 54%

the energy that is exchanged rather than supplied: 35 J of potential and 32 J of kinetic go up and down, but only 31 J of total energy has to be put in

Measured · Subject16 · 1.14 m/s · an ideal pendulum would recover 100%; walking near its best speed recovers 60-70%



Gait cycle  28%

Height only

Energy

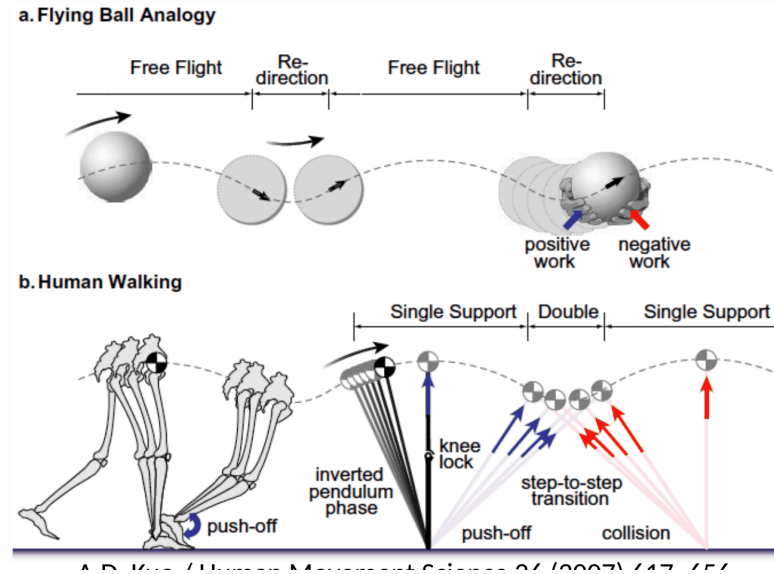
Subject16

Subject15

▶ Walk

Start with **Height only**: two arcs per stride, peaking at each midstance — the body is **highest** when it is balanced over one foot, which is the opposite of what most people guess. Then switch to **Energy**: potential and kinetic mirror each other.

Where the energy actually goes



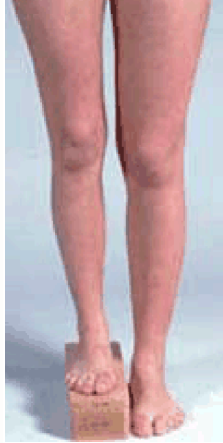
A. D. Kuo, *Human Movement Science* 26 (2007) 617–656

During double support the whole body's centre of gravity has to be **redirected** from the arc over one limb to the arc over the next.

The pendulum continues on the other side afterwards — so the major energy expenditure in walking is not the vaulting at all. It is the **redirection**.

The leading leg *collides* with the ground and takes energy out; the trailing leg pushes off and puts it back. Both cost, and the cost grows steeply with step length — which is what sets the preferred step length.

Leg length discrepancy



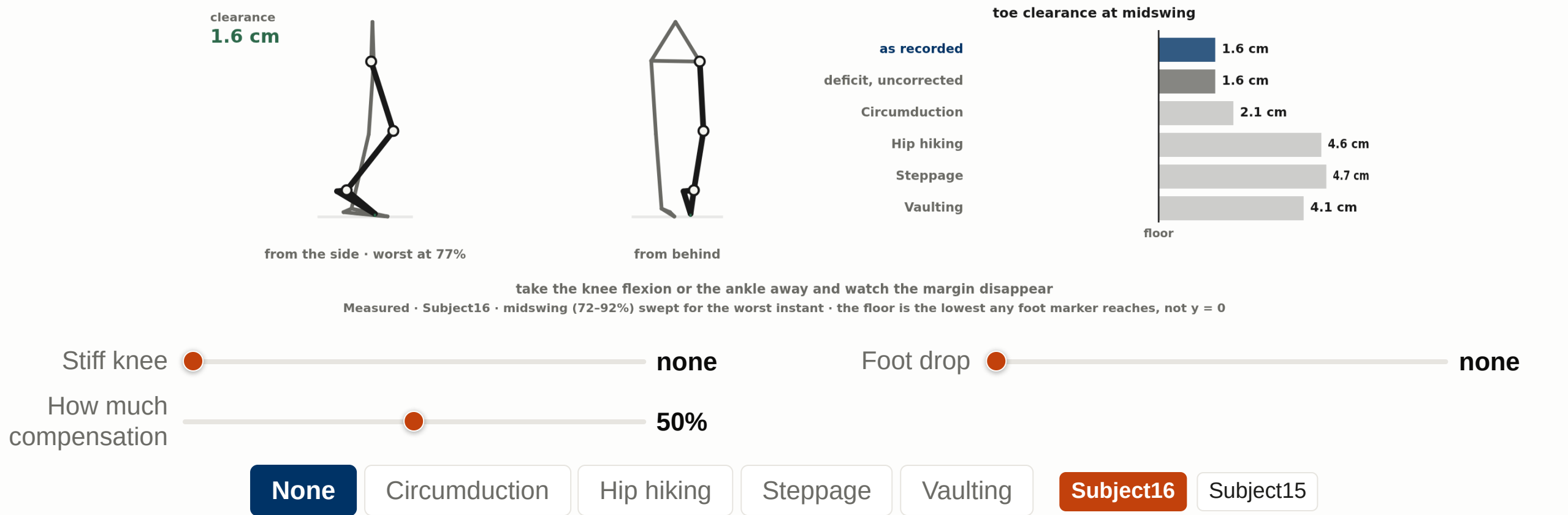
Four of the common abnormalities — circumduction, hip hiking, steppage and vaulting — are closely related: they are all compensations for the same problem, a difference in **effective** leg length between swing and stance.

Structural discrepancies are an actual anatomical shortening of one or more bones of the lower limb.

Functional leg length differences arise instead from muscular weakness or inflexibility at the pelvis, or at the foot and ankle complex.

A stiff knee or a dropped foot makes the swinging limb *functionally* too long even though the bones are the same length. The gait looks the same either way, which is why the examination has to come before the interpretation.

Buying clearance at midswing



The toe clears the floor by **a centimetre or two** — that is the whole margin. Take the knee flexion or the ankle away and it goes through. Two compensations are invisible from the side, which is why there are two views.

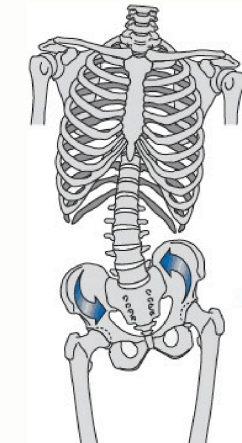
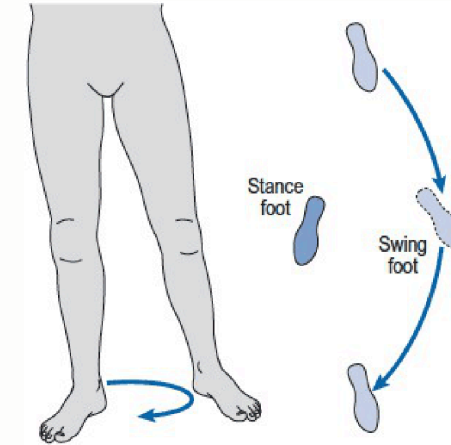
Circumduction and hip hiking

Circumduction — a swing phase modification.

- Ground contact by the swinging leg is avoided by swinging it *outward*, in an arc rather than straight forward.
- Also used to advance the limb when the hip flexors are weak, by letting the adductors act as hip flexors while the hip is extended.

Hip hiking — also a swing phase modification.

- The pelvis is lifted on the side of the swinging leg, raising the whole limb.
- Paid for by the quadratus lumborum and the trunk, on every step.



Steppage and vaulting

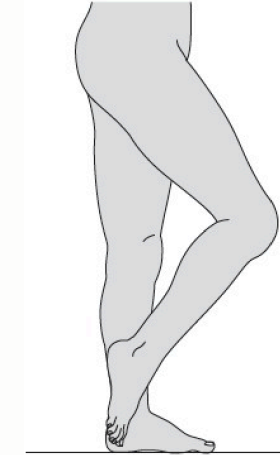
Steppage — a swing phase modification.

- Exaggerated knee and hip flexion, to lift the foot higher than usual.
- Used particularly to compensate for a plantarflexed ankle — **foot drop**, from inadequate dorsiflexion control.

Vaulting — a *stance* phase modification.

- The subject goes up on the toes of the stance leg, raising the body and so increasing the clearance for the swinging leg.
- Causes an exaggerated vertical movement of the trunk, which is wasteful of energy.

Vaulting is the odd one out, and its cost is the one the pendulum model predicts: it adds vertical excursion to the centre of mass, which is energy that has to be put in and taken out on every single step.



What to take away

- **The cycle is the unit.** Heel strike to the same heel strike; about 60% stance, 40% swing, with two periods of double support that vanish at the walk–run transition.
- **Normalise before comparing.** Stride length divided by leg length is comparable between people; stride length in centimetres is mostly a measure of height.
- **Every measurement has a definition inside it.** Step width from a heel marker is not step width from the midpoint of the heel, and the two differ by about a centimetre whatever you do to the data.
- **The vertical force has two humps because the centre of mass accelerates** — not because the effort varies. The fore–aft force brakes, then propels, and over a stride *at a steady speed* the two cancel — which a single overground stride often is not.
- **Read a joint with three panels.** Angle, moment and EMG together say concentric or eccentric and name the muscle; the moment alone says neither.
- **The support moment is the thing the body controls**, not the individual joint moments. Two healthy people make the same total in visibly different ways.
- **Walking is an inverted pendulum with a cost at each transition.** About sixty per cent of the energy is exchanged for free; the rest goes into redirecting the centre of mass twice a stride.
- **Every compensation buys something and charges for it elsewhere.**

BIOMECHANICS (EPHE 341)

Gait Analysis

Next up: angular kinetics 2 — moment of inertia, angular momentum and joint power ·
Measured data: Lencioni et al., two subjects, level walking